

9. High density lipoproteins (HDLs) and low density lipoproteins (LDLs) transport cholesterol around the body.

(a) (i) State a use of cholesterol in the body.

1

(ii) Name the organ to which HDL transports excess cholesterol for elimination.

1

(b) Familial hypercholesterolaemia (FH) is an inherited condition.

Individuals with FH have high levels of LDLs in their bloodstream.

(i) Statins are drugs, and can be prescribed to individuals with FH.

Statins act as competitive inhibitors of an enzyme involved in the synthesis of cholesterol.

Describe the mode of action of competitive inhibitors such as statins.

1

(ii) FH is caused by an autosomal dominant allele.

Describe two ways in which the patterns of inheritance in a family history would indicate autosomal dominance.

2

1. _____

2. _____

